



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 4 Report

Internship Program — Batch 2026

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Submitted To:

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1. M-Tech Production & Marketing Internship Program 2026

Daily Internship Report

Date: 16 July 2026

2. 1. Introduction

This report summarizes the activities completed during the **M-Tech Production & Marketing AI/ML Internship Program** on **16 July 2026**. During today's session, I successfully studied and completed **Experiment 12: Generative AI – GANs, VAEs & Diffusion Models** from the **Artificial Intelligence & Machine Learning Lab Manual**. The experiment focused on the rapidly growing field of **Generative Artificial Intelligence**, which enables machines to create new content such as images, text, audio, and videos. I studied different generative models to understand their architecture, learning mechanisms, and practical applications in modern AI systems.

3. 2. Objectives

The primary objectives of today's learning session were:

- To understand the concept and significance of **Generative Artificial Intelligence**.
 - To learn the working principles of **Generative Adversarial Networks (GANs)**, **Variational Autoencoders (VAEs)**, and **Diffusion Models**.
 - To compare different generative learning techniques and their characteristics.
 - To explore real-world applications of AI-generated content.
 - To understand the advantages and limitations of modern generative models.
 - To develop a foundational understanding of AI-driven content generation systems.
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4. 3. Task Description

Today's task involved studying **Experiment 12** from the AI/ML Lab Manual, which focused on **Generative AI** and its major model architectures.

The experiment introduced three important generative models:

- **Generative Adversarial Networks (GANs)**

- **Variational Autoencoders (VAEs)**
- **Diffusion Models**

I learned how each of these models learns patterns from existing datasets and generates new, realistic content. The experiment also explained practical applications of these models in **image generation, creative design, medical imaging, entertainment**, and other AI-driven multimedia systems.

5. 4. Work Done

During today's internship session, I successfully completed **Experiment 12** from the AI/ML Lab Manual.

I began by studying the fundamentals of **Generative AI** and understanding how generative models differ from traditional predictive models. Unlike predictive models that focus on classification or regression, generative models learn the underlying distribution of data and create entirely new samples that resemble the original dataset.

Next, I explored the working mechanism of **Generative Adversarial Networks (GANs)**. I learned that GANs consist of two competing neural networks: a **Generator**, which creates synthetic data, and a **Discriminator**, which evaluates whether the generated data is real or fake. Through this adversarial training process, the Generator gradually learns to produce highly realistic outputs.

I then studied **Variational Autoencoders (VAEs)** and understood how they encode data into a compressed latent representation and then reconstruct it to generate new samples with similar characteristics. This approach enables smooth generation and interpolation of data points.

Furthermore, I explored **Diffusion Models**, which generate high-quality images by starting with random noise and gradually removing the noise through multiple denoising steps until a realistic image is produced. I learned that diffusion-based models have become one of the most powerful approaches for modern AI image generation.

Finally, I examined several practical applications of Generative AI, including **AI image generation, content creation, medical image synthesis, multimedia production, virtual character generation**, and **creative design systems**. These examples demonstrated the transformative impact of Generative AI across multiple industries.

6. 5. Challenges Faced

The concepts presented in **Experiment 12** were highly advanced and intellectually engaging. However, understanding the adversarial training process of **GANs**, where two neural networks compete against each other, required additional study and careful analysis.

Comparing the architectures and learning mechanisms of **GANs**, **VAEs**, and **Diffusion Models** also required significant attention, as each model uses a fundamentally different approach for content generation. In addition, understanding how diffusion-based models gradually transform random noise into highly realistic images involved several complex mathematical and conceptual ideas.

By reviewing theoretical concepts, studying visual demonstrations, and analyzing practical examples, I developed a much clearer understanding of these modern generative techniques.

7. 6. Key Learnings

During today's learning session, I achieved the following outcomes:

- Developed a strong understanding of the fundamentals of **Generative Artificial Intelligence**.
- Learned how **GANs** use Generator and Discriminator networks to create realistic synthetic data.
- Understood how **VAEs** compress and reconstruct data through latent representations.
- Learned how **Diffusion Models** generate high-quality images through gradual denoising.
- Explored the strengths, limitations, and use cases of different generative model architectures.
- Understood the growing importance of Generative AI in **design, healthcare, entertainment, digital media, and content creation**.
- Strengthened my understanding of modern AI technologies used for creative and synthetic data generation.

8. 7. Conclusion

The successful completion of **Experiment 12** provided valuable theoretical knowledge and practical insight into **Generative Artificial Intelligence** and its major model architectures. I developed a comprehensive understanding of how **GANs**, **VAEs**, and **Diffusion Models** learn from existing data and generate realistic new content.

The knowledge and practical skills gained during today's session have significantly enhanced my understanding of modern AI content generation technologies and their real-world applications. This experiment has provided a strong foundation for exploring advanced Generative AI systems and understanding their transformative role in future Artificial Intelligence and Machine Learning applications.

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12. Intern Information

Field	Details
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Intern Name	Abdullah Rafique
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Supervisor Name	Mr. Mohazin Zahid
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Date	16 July, 2026
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Field	Details
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Domain	AI/ML Intern
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Intern Signature	<u>ABD</u>
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